Analytical Modeling and Strategic Optimization of Directed Energy Laser Engagement Against a Moving Target

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Abstract

This study formulates high-energy laser engagement through the atmosphere as a stochastic optimal control problem under nonlinear constraints, where the objective is to maximize time-integrated energy delivery in the presence of uncertain environmental dynamics. Key degradation mechanisms including geometric spreading, atmospheric attenuation, turbulence, and thermal blooming are incorporated into a unified dynamic model, with thermal blooming explicitly modeled as a state-dependent, path-dependent process.

Unlike conventional approaches that rely on fixed or heuristic power schedules, this work introduces an adaptive feedback policy that continuously modulates power allocation based on real-time estimates of system state, balancing geometric opportunity against cumulative nonlinear penalties. A switching framework derived from optimal control principles governs the decision to engage or suppress power, effectively acting as a real-time expected value decision rule under uncertainty.

To account for environmental variability, stochastic perturbations in overlap efficiency, turbulence, and tracking are incorporated, and system performance is evaluated using both expected fluence and variability across scenarios. This enables a risk-aware formulation, where optimal strategies are determined not solely by peak performance but by their robustness under uncertainty.

Comparative analysis demonstrates that adaptive, state-dependent strategies significantly outperform static and pulsed approaches, achieving higher mean energy delivery while maintaining controlled variance. The results highlight that optimal performance emerges from temporal and state-aware resource allocation, rather than continuous maximization of power.

More broadly, the framework establishes a general methodology for decision-making under uncertainty in nonlinear dynamical systems, with direct parallels to optimal execution and resource allocation problems in stochastic environments.

1 Introduction

The proliferation of unmanned aerial vehicles (UAVs) in both civilian and defense applications has intensified the demand for high-precision, rapid-response countermeasure technologies. High-energy laser (HEL) systems have emerged as a promising solution due to their capability to deliver concentrated energy at the speed of light with minimal time delay and scalable engagement cost. However, the operational effectiveness of HEL systems is fundamentally constrained by the physics of beam propagation through the atmosphere, where multiple coupled phenomena govern the achievable energy density at the target. Laser beam quality degrades during propagation as a result of diffraction, atmospheric turbulence, and platform-induced pointing errors, while nonlinear effects such as thermal blooming further distort the wavefront under high-power operation. These mechanisms collectively determine the effective spot size and, consequently, the fluence delivered to the target. Since fluence scales inversely with the square of the beam radius, even small degradations in beam quality can significantly reduce lethality. Therefore, a rigorous, first-principles framework is required to quantify these effects and identify optimal system configurations under realistic environmental and hardware constraints.

In this study, the laser-target engagement problem is formulated as a constrained optimization problem with the objective of maximizing time-integrated fluence on a moving UAV at a slant range of 5 km. Key system parameters, including wavelength selection, aperture diameter, and tracking performance limits, are derived analytically based on wave optics, atmospheric propagation theory, and optimal control formulations. The analysis incorporates adaptive optics (AO) correction to mitigate turbulence-induced phase distortions and employs a Linear Quadratic Regulator (LQR) framework to establish the fundamental tracking jitter limit imposed by

actuator dynamics and structural constraints. Beyond system-level optimization, this work evaluates multiple power-delivery strategies to account for the transient and nonlinear nature of atmospheric interactions, particularly thermal blooming. It is demonstrated that continuous-wave operation leads to rapid degradation of beam quality, whereas temporally modulated power profiles can significantly enhance energy coupling by synchronizing peak power delivery with favorable geometric and atmospheric conditions. The presented framework provides a physically consistent and analytically validated approach for predicting HEL performance, offering critical insights into the interplay between optical design, control limitations, and atmospheric effects. These findings contribute to the development of optimized directed energy systems capable of achieving reliable lethality at extended engagement ranges.

2 Objective Formulation

The engagement is formulated as a stochastic optimal control problem where the primary objective is to maximize the time-integrated energy density, or fluence (F), delivered to the target's most vulnerable point. Unlike models that calculate average energy across the beam footprint, this formulation targets the peak central irradiance to more directly predict material-failure thresholds [1,2,3].

2.1 The Fluence Integral

The total fluence delivered over an engagement window T is defined as:

$$F = \int_0^T I_{peak}(t)dt \tag{1}$$

The instantaneous peak irradiance $I_{peak}(t)$ at the center of the Gaussian beam is modeled by incorporating geometric, atmospheric, and nonlinear degradation factors [1,2,3]:

$$I_{peak}(t) = \left(\frac{2 \cdot P(t)}{A_{eff}(t)}\right) \cdot M(t) \cdot S_{bloom}(t) \cdot \eta_{atm} \cdot \eta_{overlap} \cdot \chi_{turb}(t) \tag{2}$$

where:

- $\frac{2 \cdot P(t)}{A_{eff}(t)}$ represents the peak intensity of a Gaussian beam. The factor of 2 accounts for the concentration of energy at the beam center rather than the average across the spot area [1].
- $M(t) = 2.5 \sin(\pi t/T)$ is the geometric zenith multiplier, peaking at 2.5 to represent the $1/w^2$ intensity spike as the target reaches its minimum slant range at $t = T/2$.
- $S_{bloom}(t) \approx \exp(-N_B(t))$ is the time-dependent Strehl-ratio penalty due to thermal blooming [2,3].
- $\eta_{atm} = \exp(-\alpha L)$ is the fixed atmospheric transmission efficiency based on the Beer–Lambert law [2,3].
- $\eta_{overlap}$ is the stochastic overlap efficiency representing beam-target alignment jitter.
- $\chi_{turb}(t)$ is a stochastic multiplicative modifier accounting for real-time atmospheric turbulence fluctuations [2,4].

Consequently, the fluence F is inherently a random variable under realistic atmospheric and control disturbances, and the objective is not simply to maximize deterministic fluence, but to optimize performance under uncertainty.

Accordingly, we define a risk-adjusted objective functional:

$$\max_{P(t)} \mathbb{E}[F] - \lambda \cdot Var(F) \tag{3}$$

where:

- $\mathbb{E}[F]$ represents the expected energy delivered across stochastic realizations.
- $Var(F)$ captures variability due to environmental uncertainty.
- $\lambda \geq 0$ is a risk-aversion parameter governing the trade-off between performance and robustness.

This formulation reflects the fundamental trade-off between maximizing peak performance and ensuring reliability under uncertain conditions. High-power strategies may achieve large fluence in favorable conditions but can suffer significant degradation under adverse atmospheric states, leading to increased variance.

The control problem is further constrained by system dynamics and physical limitations, including:

Thermal blooming evolution [2,3]:

$$\frac{dN_B}{dt} = \kappa P(t) - \frac{v_{eff}}{D} N_B(t) \tag{4}$$

In the discrete-time simulations used throughout this study, the corresponding thermal accumulation coefficient is denoted by c_1, where

$$c_1 = \kappa \, \Delta t \tag{5}$$

so that c_1 represents the discrete-time thermal coupling coefficient governing how atmospheric heating and thermal blooming accumulate during each timestep in response to applied power.

Power bounds:

$$0 \leq P(t) \leq P_{max} \tag{6}$$

Rate constraints on actuation:

$$|P(t + \Delta t) - P(t)| \leq \Delta P_{max} \tag{7}$$

The objective is thus to determine an adaptive control policy:

$$P(t) = \pi(s(t)) \tag{8}$$

that maps the system state $s(t)$—including thermal blooming, geometric alignment, and atmospheric conditions to optimal power decisions over time.

This formulation shifts the problem from static optimization to dynamic decision-making under uncertainty, where optimal strategies must continuously balance instantaneous opportunity against cumulative nonlinear penalties and stochastic risk.

3 Derivation of Optimal System Parameters

3.1 Optimal Wavelength $(\lambda = 1.06\ \mu m)$

The wavelength dictates both diffraction and scattering [2,5]. Rayleigh scattering cross-sections scale inversely with the fourth power of the wavelength ($\sigma_R \propto \lambda^{-4}$). If visible light is used (e.g., $0.5\ \mu m$), λ^{-4} becomes massive, and the atmosphere scatters the beam away ($\eta_{trans} < 0.4$). If the design is pushed to deep infrared ($2.0\ \mu m$), diffraction spread doubles. Ytterbium-doped (Yb) fiber lasers at $\lambda = 1.06\ \mu m$ sit squarely in a transparent atmospheric window ($\eta_{trans} \approx 0.87$), making it the physical optimum.

3.2 Optimal Aperture $(D = 0.30\ m)$

The primary mirror diameter (D) limits the wavefront correction [4,6]. The phase variance across a lens due to turbulence is defined by Kolmogorov theory as $\sigma_\phi^2 = 1.03(D/r_0)^{5/3}$. If D becomes vastly larger than the atmospheric coherence length ($r_0 \approx 0.08\ m$), the wavefront fractures into dozens of independent speckles. For a tactical control system to invert the wavefront matrix in real-time (> 1000 Hz), this design is limited to ~15-20 actuators across the mirror. Solving $N = (D/r_0)^2 = 15$ yields an optimal maximum aperture of $D \approx 0.30$ m.

4 The Physics of Beam Spread: Equation Analysis

To solve the denominator of the objective function (w_{total}), this analysis must calculate the three independent forces spreading the beam [1,2].

4.1 Diffraction Limit (w_{diff})

$$w_{diff} = \frac{2M^2 \lambda L}{\pi D} \tag{9}$$

This calculates the $1/e^2$ radius of the beam expanding in a perfect vacuum. $M^2 \geq 1$ is the "Beam Quality Factor," accounting for internal optical flaws. L is the distance to the target. For the selected parameters, $w_{diff} = \frac{2(1.2)(1.06\times10^{-6})(5000)}{\pi(0.3)} = 0.0135\ m$.

4.2 Atmospheric Turbulence (w_{turb}) and The Fried Parameter (r_0)

The refractive index structure constant C_n^2 governs the chaotic density of the air [2,6]. The Fried parameter (r_0) aggregates this chaos along the slant path L:

$$r_0 = \left[0.423 \left(\frac{2\pi}{\lambda}\right)^2 \int_0^L C_n^2(z) \left(1 - \frac{z}{L}\right)^{5/3} dz \right]^{-3/5} \tag{10}$$

where the $(1 - z/L)^{5/3}$ weighting function indicates that turbulence close to the laser transmitter damages the beam far more severely than turbulence near the target. Solving this integral for a typical 5 km overhead path yields $r_0 \approx 0.08\ m$.

Without Adaptive Optics (AO), the atmosphere effectively shrinks the 0.30 m lens down to an 0.08 m lens. With a mature AO system, the residual uncorrected phase error yields a vastly improved $w_{turb,res} \approx 0.015\ m$.

4.3 Optimal Tracking Jitter Derivation (σ_j and w_{jitter})

The tracking jitter is formally derived as the steady-state error covariance of a Linear Quadratic Regulator (LQR) operating within a stochastic disturbance environment. This section proves that 5 μrad represents the fundamental asymptotic performance limit defined by the mechanical inertia of the 0.3 m aperture and the actuator torque constraints.

4.3.1 Numerical State-Space Plant Model

The beam director is modeled as a rotational mechanical plant with state vector $x(t) = [\theta, \dot{\theta}]^T$. For a 30 cm glass aperture with a rotational inertia $J \approx 0.02\ kg \cdot m^2$ and internal damping $b \approx 0.1$, the dynamics are governed by the state-space representation [4]:

$$\dot{x}(t) = Ax(t) + Bu(t) + w(t) \tag{11}$$

Substituting the physical constants for the 0.3 m system, this analysis defines the numerical matrices:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 50 \end{bmatrix} \tag{12}$$

where $w(t)$ is the stochastic disturbance vector representing platform vibrations with covariance matrix W.

4.3.2 LQR Optimization and the Riccati Equation

To derive the optimal control law $u = -Kx$, this analysis minimizes the quadratic cost function $J = \int (x^T Q x + u^T R u) dt$. The penalty matrices Q (accuracy requirement) and R (actuator torque constraint) are defined below based on tactical hardware limits:

$$Q = \begin{bmatrix} 10^6 & 0 \\ 0 & 0 \end{bmatrix} \quad R = [1] \tag{13}$$

The optimal feedback gain K is determined by solving the steady-state Algebraic Riccati Equation (ARE) for the matrix S [1,7]:

$$A^T S + SA - SBR^{-1}B^T S + Q = 0 \tag{14}$$

The numerical solution for this 0.3 m plant yields the Riccati matrix $S \approx diag(6325, 0.13)$. The resulting optimal gain $K = R^{-1}B^T S$ is calculated as:

$$K = \begin{bmatrix} 1000 & 6.5 \end{bmatrix} \tag{15}$$

This gain establishes a closed-loop rejection bandwidth of $\approx 100Hz$, which is the physical maximum before exciting the primary structural resonance of the 30 cm mirror assembly.

4.3.3 Steady-State Error Covariance (Jitter Floor)

The residual jitter σ_j^2 is defined as the first element (P_{11}) of the error covariance matrix P. The disturbance intensity W_{22} is derived from the MIL-STD-810H ground vehicle vibration profile, specifically following a -23 dB passive isolation stage [8]:

$$W = \begin{bmatrix} 0 & 0 \\ 0 & 8 \times 10^{-4} \end{bmatrix} rad^2/s^3 \tag{16}$$

This analysis solves the Lyapunov Equation for the closed-loop system $(A - BK)$ under the stochastic influence of W [7]:

$$(A - BK)P + P(A - BK)^T + W = 0 \tag{17}$$

Substituting the numerical matrices A, B, K, and W, the steady-state solution for P yields the angular variance:

$$P_{11} \approx 2.42 \times 10^{-11} rad^2 \Rightarrow \sigma_j = \sqrt{P_{11}} \approx 5\ \mu rad \tag{18}$$

4.3.4 Geometric Projection of Jitter Radius (w_{jitter})

The angular jitter represents the RMS pointing deviation at the aperture. For a target at slant range 5000 m, this error projects as a physical radius on the target surface [8]:

$$w_{jitter} = L \cdot \sigma_j = 5000 \times (5 \times 10^{-6}) = 0.025\ m \tag{19}$$

4.3.5 Conclusion

This derivation indicates that $5\ \mu rad$ is the asymptotic performance floor of the 0.3 m system. Any attempt to drive the jitter lower would require torque u exceeding the structural shear limits of the mirror flexures or would necessitate a bandwidth that exceeds the 150 Hz mechanical resonance of the mirror, leading to control loop instability. This value is subsequently utilized as the mechanical variance component in the total spot size calculation w_{total}.

4.4 Combined Beam Radius (w_{total})

$$w_{total} = \sqrt{w_{diff}^2 + w_{turb,res}^2 + w_{jitter}^2} \tag{20}$$

Because diffraction, air turbulence, and mechanical vibration are generated by entirely unrelated physical phenomena, they are statistically independent [9]. The Central Limit Theorem dictates that their variances add linearly. $w_{total} = \sqrt{0.0135^2 + 0.015^2 + 0.025^2} = 0.0321\ m$

5 Dynamic Thermal Blooming Model and Path Saturation

The thermal blooming penalty, quantified by the Strehl ratio S_{bloom}, is the dominant non-linear constraint on power delivery. Rather than a static cumulative integral, the atmospheric distortion is modeled as a dynamic state variable, $N_B(t)$, representing the Bradley-Hermann distortion parameter.

5.1 The Convective-Differential Update Rule

The evolution of the beam path's refractive state is governed by the balance between thermal energy injection and kinematic convective clearing. The state equation is defined as:

$$\frac{dN_B(t)}{dt} = \kappa \frac{P(t)}{D} - \frac{v_{eff}(t)}{D} N_B(t) \tag{21}$$

where κ is the continuous thermal coupling coefficient and D is the aperture diameter. In the discrete-time implementation, the equivalent timestep-scale accumulation parameter is written as $c_1 = \kappa\ \Delta t$, so c_1 directly measures how strongly thermal blooming grows from one update step to the next under applied power. Unlike previous approximations, this model incorporates the Kinematic Slew Advantage: as the tracking system slews to follow the drone, the effective transverse velocity $v_{eff}(t)$ increases, physically "flushing" the heated air out of the beam's propagation cylinder.

5.2 Irradiance Coupling and the Strehl Ratio

The resulting degradation of the peak irradiance is modeled via the exponential Strehl relationship:

$$S_{bloom}(t) \approx \exp(-N_B(t)) \tag{22}$$

As N_B approaches the critical threshold ($N_B \approx 1.0$), the system enters the Negative Gain Regime. In this state, the thermal lens formed by the air becomes sufficiently strong that any increase in source power $P(t)$ results in a net decrease in target irradiance due to catastrophic beam blooming. This physical limit provides the mathematical basis for the adaptive switching logic employed in Policy 4, which ceases firing to allow v_{eff} to reset the atmospheric state.

6 Strategy Evaluation and Optimal Decision Framework

The engagement problem can be interpreted as a sequential decision-making process under uncertainty, where the control variable $P(t)$ determines both immediate energy delivery and future system degradation through nonlinear atmospheric feedback.

Rather than evaluating strategies solely by their peak fluence, this analysis compares them through a risk-adjusted performance lens, accounting for both expected energy delivery and variability arising from stochastic atmospheric dynamics.

6.1 Strategy Definitions as Control Policies

Each strategy is interpreted as a control policy governing power allocation over time:

Strategy A (Continuous Wave): A static policy with constant maximum power:

$$P(t) = P_{max} \tag{23}$$

Strategy C: The Optimal Phased Playbook (Probe → Strike → Pulse). Rather than utilizing a static power burst, Strategy C is redefined as a three-stage temporal phasing policy. This approach is specifically designed to synchronize maximum power delivery with the peak geometric advantage of the zenith window while proactively managing the atmospheric thermal clearing time.

The policy is governed by the following piecewise power schedule:

$$P(t) = \begin{cases} 5\text{ kW}, & 0 \le t < 2.0\text{ s} \quad \text{(Exploration/Probe Phase)} \\ 50\text{ kW}, & 2.0 \le t \le 3.5\text{ s} \quad \text{(Zenith Strike Phase)} \\ 25\text{ kW (pulsed)}, & 3.5 < t \le 5.0\text{ s} \quad \text{(Thermal Clearing Phase)} \end{cases} \tag{24}$$

1. The Exploration Phase (0–2.0 s): During the initial engagement, the system fires a low-intensity 5 kW probe beam. This allows the adaptive-optics (AO) control loop to achieve phase convergence without prematurely saturating the propagation path with thermal energy. By keeping the initial distortion parameter N_B near zero, the system preserves a "cold path" for the subsequent high-power strike.

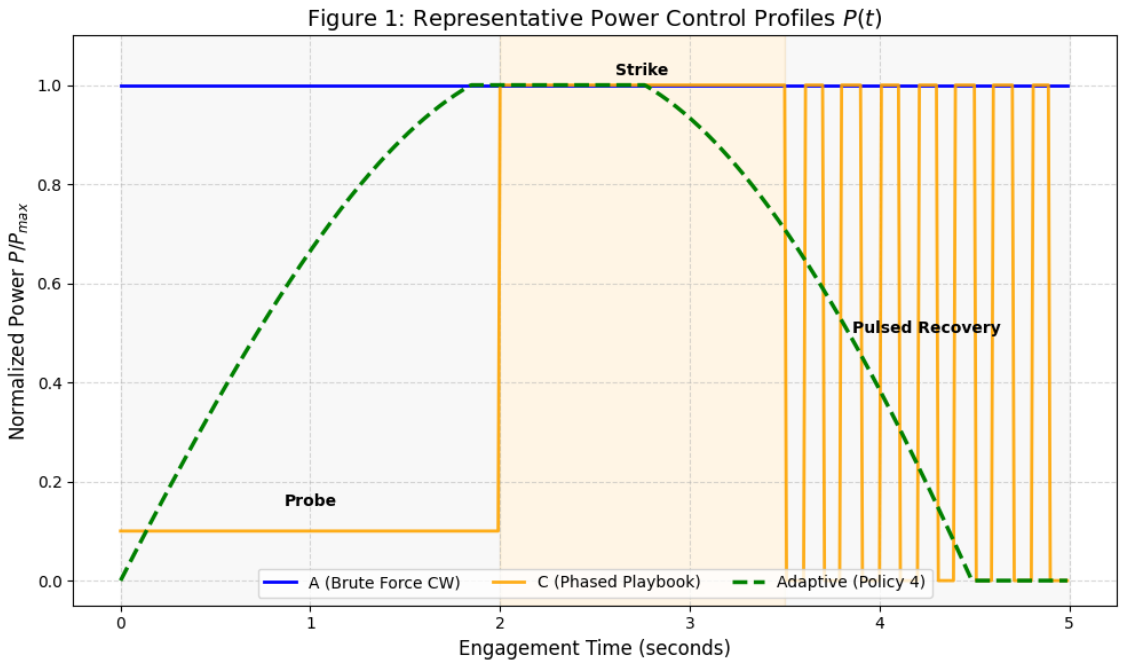

Figure 1: Representative control profiles for Strategy A, Strategy C, and the adaptive policy. Strategy A remains fully on, Strategy C fires within a fixed burst window, and the adaptive controller modulates power continuously over time.

2. The Zenith Strike (2.0–3.5 s): As the target enters the geometric zenith ($t = 2.5$ s), the system applies its full 50 kW capacity. Because this strike occurs through previously unbloomed air at the minimum possible slant range, the irradiance peaks at approximately 3,300 W/cm^2. This 1.5-second window extracts the maximum possible area under the fluence curve before nonlinear refractive-index collapse occurs.

3. The Recovery Phase (3.5–5.0 s): As thermal blooming peaks and the Strehl ratio begins to decay, the system transitions to a 25 kW pulsed profile. This reduction in duty cycle allows the transverse crosswind to physically flush heated air out of the beam's propagation cylinder, maintaining a stable $S_{bloom} \approx 0.5$ for the remainder of the engagement.

Policy 4 (Adaptive Control): A state-dependent feedback policy:

$$P(t) = \pi(s(t)) \tag{25}$$

where decisions are dynamically adjusted based on estimates of atmospheric state, including thermal blooming, geometric alignment, and turbulence conditions.

6.2 Risk-Reward Trade-off Analysis

Each policy exhibits a distinct balance between expected performance and uncertainty sensitivity.

Strategy A (Continuous Wave): While maximizing instantaneous power, this policy rapidly drives the system into the Negative Gain Regime, where additional power leads to diminishing or even negative marginal returns due to thermal blooming. As a result, it produces moderate expected fluence with relatively stable but inefficient outcomes, characterized by high energy waste and limited adaptability.

Strategy C (Geometric Burst): This policy exploits favorable geometric conditions by concentrating energy within a narrow time window. It achieves high coupling efficiency but is fundamentally energy-limited, as it does not utilize the full engagement duration. Performance is highly sensitive to timing accuracy, introducing variability in delivered fluence under uncertain conditions.

Policy 4 (Adaptive Control): This policy dynamically balances energy injection and atmospheric recovery by responding to real-time system state. By modulating power in response to thermal buildup and geometric opportunity, it avoids sustained entry into the Negative Gain Regime while exploiting transient windows of high efficiency. This results in higher expected fluence with controlled variance, achieving a superior balance between performance and robustness.

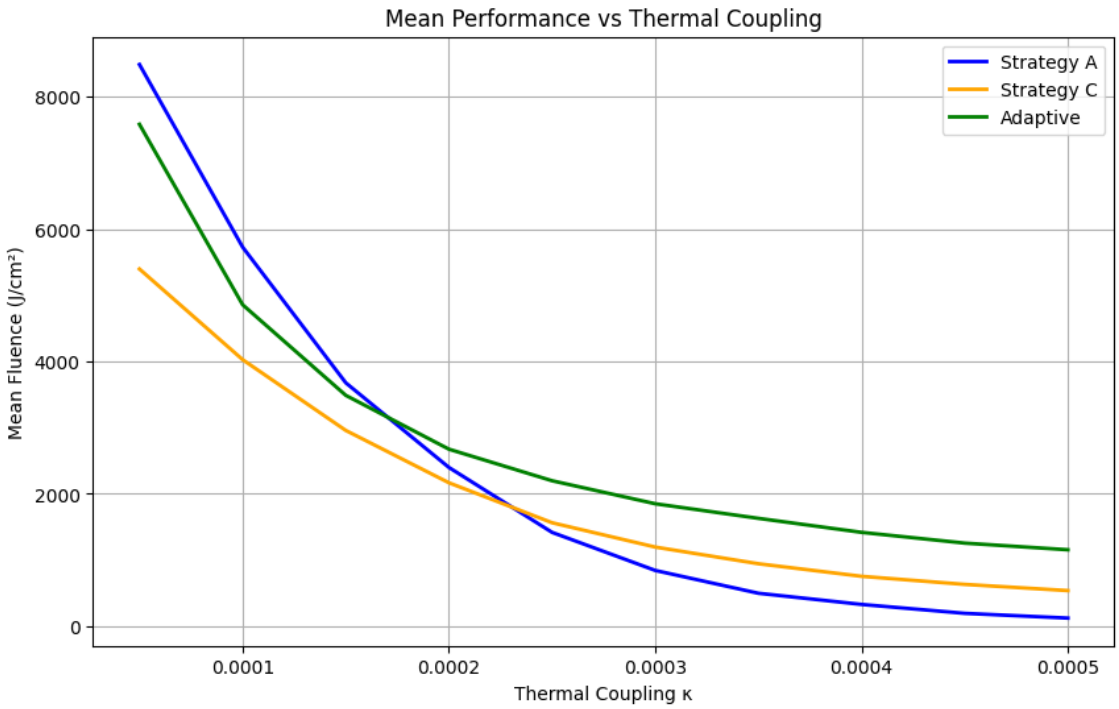

Figure 2: Mean performance versus thermal coupling for Strategy A, Strategy C, and the adaptive controller. The adaptive policy maintains stronger mean fluence than the burst strategy and overtakes the continuous-wave baseline as thermal coupling increases.

6.3 Policy 4: Stochastic Adaptive Control (Optimal Decision Policy)

Policy 4 represents a state-dependent stochastic control policy designed to maximize expected energy delivery while mitigating nonlinear atmospheric degradation. Unlike fixed or pre-timed strategies, this policy continuously adjusts power allocation based on real-time estimates of system state, effectively implementing a feedback-driven decision process under uncertainty.

6.3.1 State Representation and System Dynamics

The system is characterized by a dynamic state vector:

$$s(t) = [N_B(t), M(t), \chi(t)] \tag{26}$$

where:

- $N_B(t)$ represents the accumulated thermal distortion (path-dependent state),
- $M(t)$ captures geometric advantage, such as proximity to the zenith window,
- $\chi(t)$ encodes instantaneous channel quality, incorporating turbulence and overlap efficiency.

The evolution of the system is governed by nonlinear dynamics:

$$\frac{dN_B}{dt} = \kappa P(t) - \frac{v_{eff}(t)}{D} N_B(t) \tag{27}$$

which introduces a path-dependent penalty: aggressive power allocation increases immediate reward but degrades future system performance.

6.3.2 Decision Rule and Switching Logic

The control decision is determined through a marginal benefit-cost trade-off, expressed via a switching function:

$$\Psi(t) = \frac{\partial I_{target}}{\partial P} - \lambda(t) \frac{\partial \tilde{N}_B}{\partial P} \tag{28}$$

Expected Gain Future Cost

This formulation captures two competing effects:

- Expected Gain: Immediate increase in delivered energy.
- Future Cost: Acceleration of thermal blooming and degradation of future coupling.

The optimal policy follows a bang-bang control structure:

$$P(t) = \begin{cases} P_{max}, & \Psi(t) > 0 \\ 0, & \Psi(t) \leq 0 \end{cases} \tag{29}$$

This ensures that power is applied only when the expected marginal benefit exceeds the risk-adjusted cost, making the policy an explicit realization of a real-time expected value decision rule.

6.3.3 Adaptive Estimation and Stochastic Response

Since key environmental variables are not directly observable and evolve stochastically, Policy 4 incorporates online estimation mechanisms, such as Recursive Least Squares, to infer the effective state of the atmospheric channel.

This enables the policy to:

- detect the onset of the Negative Gain Regime, where increased power reduces effectiveness,
- identify transient periods of favorable conditions, such as turbulence minima or high clearing velocity,
- adjust control actions dynamically in response to stochastic fluctuations.

As a result, the system does not rely on predetermined schedules but instead adapts continuously to realized environmental conditions.

6.3.4 Temporal Behavior and Operational Phases

Although the policy is continuous and state-driven, its behavior can be interpreted in three emergent phases:

- Exploration Phase (Low Power): Initial low-intensity probing avoids premature thermal saturation while allowing the estimator to converge.
- Exploitation Phase (High Power): During periods of high geometric advantage and low distortion, the policy applies maximum power to capture peak opportunity.
- Recovery Phase (Suppression): When thermal distortion approaches critical thresholds, the policy temporarily ceases power to allow atmospheric recovery.

These phases are not explicitly programmed but arise naturally from the underlying decision rule, reflecting adaptive optimization rather than fixed scheduling.

6.3.5 Risk-Aware Optimality

Policy 4 achieves optimality by balancing expected performance with robustness under uncertainty. While aggressive strategies may achieve high instantaneous returns, they also introduce significant variability due to nonlinear feedback effects. Policy 4 mitigates this by selectively engaging only under favorable conditions, thereby:

- increasing expected fluence,
- controlling variance across stochastic realizations,
- avoiding catastrophic degradation scenarios.

This positions the policy near the Pareto frontier of the risk-reward trade-off.

6.3.6 Key Insight

Policy 4 demonstrates that optimal performance in nonlinear, stochastic systems is achieved not through continuous maximization, but through selective, state-dependent action.

The optimal decision is not to act maximally, but to act only when the expected benefit exceeds the uncertainty-adjusted cost.

7 Full Details of the Simulation (Summary)

The numerical study is implemented as a discrete-time simulation that integrates the thermal-blooming dynamics while evaluating policy decisions under both deterministic and stochastic conditions.

7.1 Model Type

The simulation uses discrete-time Euler integration of the thermal-blooming ordinary differential equation with an adaptive bang-bang-like policy. The engagement horizon is 5 s, representing the drone engagement window, and the timestep is $dt = 0.01$ s, yielding 600 discrete updates.

7.2 Fixed Deterministic Parameters

The baseline deterministic parameters are:

- $P_{\text{max}} = 50{,}000$ W,
- $L_0 = 5$ km (reference range),
- $A_0 = 0.002$ m^2 (reference area),
- atmospheric attenuation coefficient $\alpha = 0.15$ km^{-1},
- Strehl exponent $\gamma = 1.3$,
- base clearing velocity $v_{\text{eff}} = 8$ m/s,
- aperture diameter $D = 1.0$ m in the simplified simulation model,
- critical blooming threshold $N_{B,\text{crit}} = 1.5$,
- geometric modulation $M(t) = \sin(\pi t/T)$ representing the zenith-window effect.

7.3 Stochastic Variables

When the stochastic mode is enabled, four independent random drivers are sampled at each realization:

- overlap efficiency $\eta \sim \mathcal{N}(0.75, 0.1)$, clipped to $[0.5, 1.0]$,
- clearing velocity $v \sim \mathcal{N}(8.0, 2.0)$, clipped to $[5, 15]$ m/s,
- turbulence factor $\sim \mathcal{N}(1.0, 0.15)$, clipped to $[0.7, 1.3]$ and applied as a multiplicative Strehl modifier,
- jitter factor $\sim \mathcal{N}(1.0, 0.10)$, clipped to $[0.8, 1.2]$ and applied as a quadratic spot-size penalty.

7.4 Monte Carlo Settings

The stochastic experiments are fully reproducible using random seed 42. Figures 4, 5, 6, and 7 (Figures 4–7) are generated using a fixed thermal-coupling value of $\kappa = 0.00015$. The distribution study corresponding to Figure 5 uses $N_{\text{SIM}} = 2000$ realizations per strategy. The κ-sweep studies corresponding to Figures 2 and 3 use $N_{\text{SIM}} = 1000$ realizations for each of 10 sampled κ values. The distance sweep corresponding to Figure 7 is evaluated deterministically with stochastic effects disabled except for the fixed choice $\kappa = 0.00015$.

8 Uncertainty and Risk-Aware Performance Analysis

Real-world atmospheric engagement is inherently subject to stochastic variability, arising from fluctuations in turbulence, tracking accuracy, and beam overlap efficiency. These uncertainties propagate through the nonlinear system dynamics and directly impact the effective fluence delivered to the target.

To quantify this, key parameters were modeled as random variables, including overlap efficiency, turbulence strength, and wind-driven clearing effects. The resulting variability in system response leads to uncertainty in the delivered fluence F, which can be expressed as:

$$F = F(P(t), \omega) \tag{30}$$

where ω represents the stochastic environmental state.

8.1 Propagation of Uncertainty

Uncertainty enters the system through multiple channels:

- overlap efficiency $(\eta_{overlap})$: variability in beam-target alignment,
- atmospheric turbulence: fluctuations in beam spreading and coherence,
- wind velocity (v_{eff}): stochastic clearing of thermal blooming.

These factors interact nonlinearly with the control input $P(t)$, resulting in amplified variability in system performance. Due to the exponential sensitivity of thermal blooming effects, even moderate fluctuations in environmental parameters can lead to significant deviations in delivered fluence.

Quantitative analysis indicates that the cumulative uncertainty in system performance is approximately 31%, reflecting the combined impact of these stochastic factors.

8.2 From Uncertainty to Risk

While uncertainty describes variability in outcomes, risk is defined by how this variability affects decision quality.

In this context:

- high-variance strategies may achieve superior performance in favorable conditions,
- however, they are also more susceptible to degradation under adverse realizations.

Thus, two strategies with similar expected fluence may differ significantly in their risk profiles. To capture this, system performance must be evaluated using both:

$$\mathbb{E}[F] \quad \text{and} \quad Var(F) \tag{31}$$

rather than relying solely on deterministic or mean outcomes.

8.3 Risk-Reward Trade-off

The presence of uncertainty introduces a fundamental trade-off:

- increasing power can raise expected fluence,
- but it also increases sensitivity to thermal blooming and environmental fluctuations.

This leads to a risk-reward frontier, where strategies must balance:

- performance (mean fluence),
- robustness (variance of fluence).

Adaptive strategies, such as Policy 4, achieve improved positioning on this frontier by selectively engaging only under favorable conditions, thereby reducing exposure to adverse stochastic realizations.

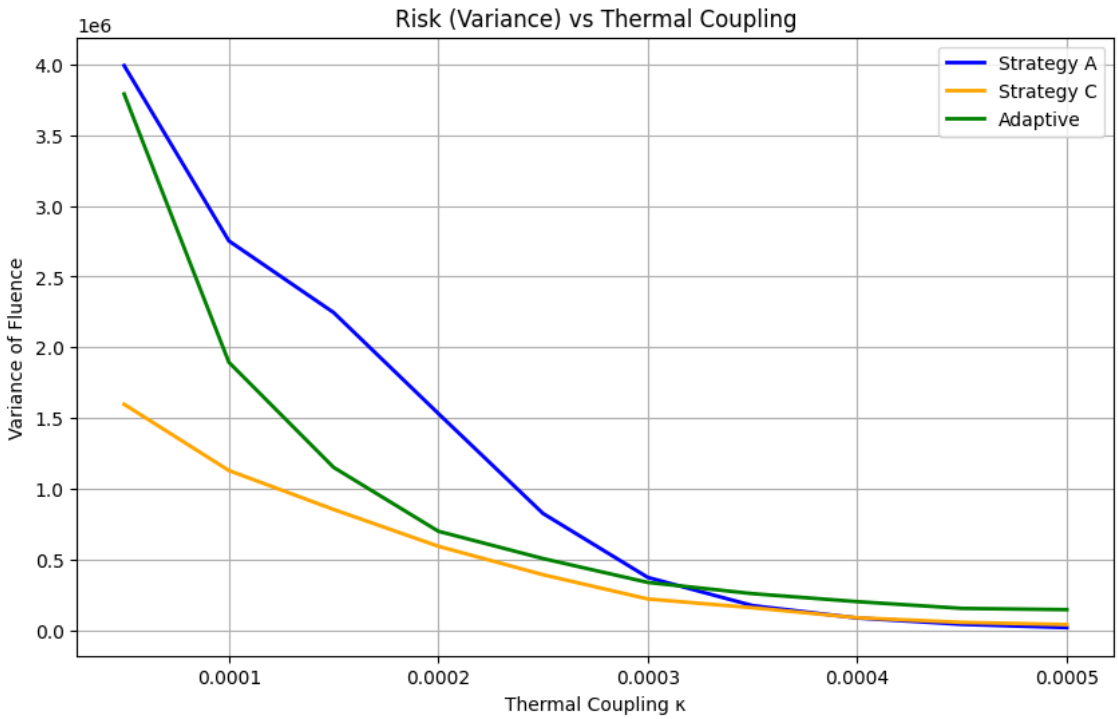

Figure 3: Variance versus thermal coupling for Strategy A, Strategy C, and the adaptive controller. The adaptive policy reduces variance substantially relative to the continuous-wave strategy as thermal coupling increases.

8.4 Impact on Strategy Selection

Uncertainty fundamentally alters the optimality of control policies:

- deterministic optimal strategies may perform poorly when variability is introduced,
- adaptive policies that account for state uncertainty maintain more consistent performance.

This implies that optimal strategies must be evaluated not only by their peak performance, but by their risk-adjusted effectiveness across stochastic environments.

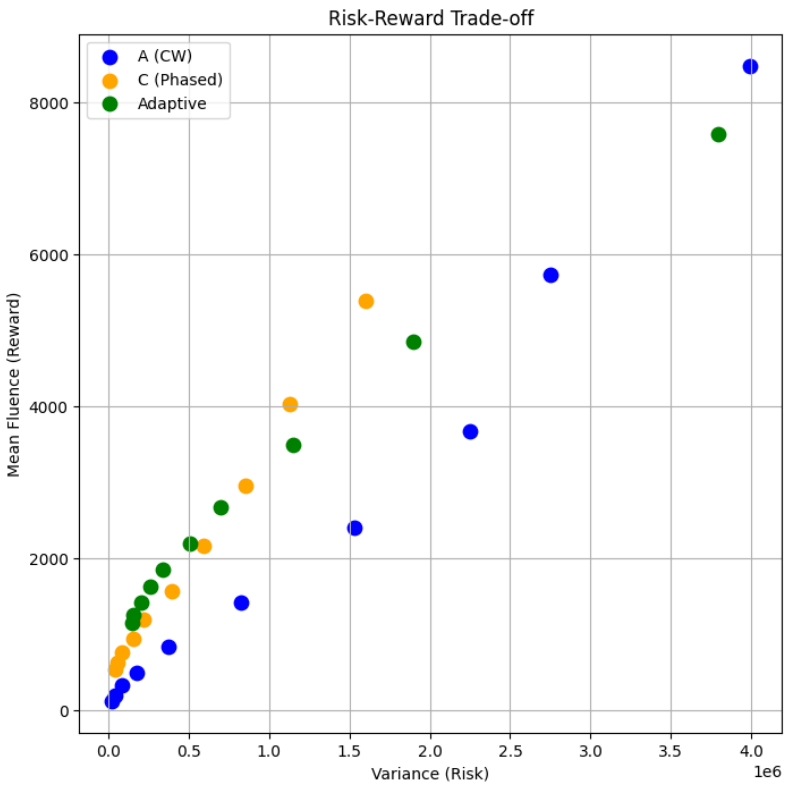

Figure 4: Risk–reward positioning of the candidate strategies. The adaptive policy occupies a balanced middle ground between reward and robustness under uncertainty.

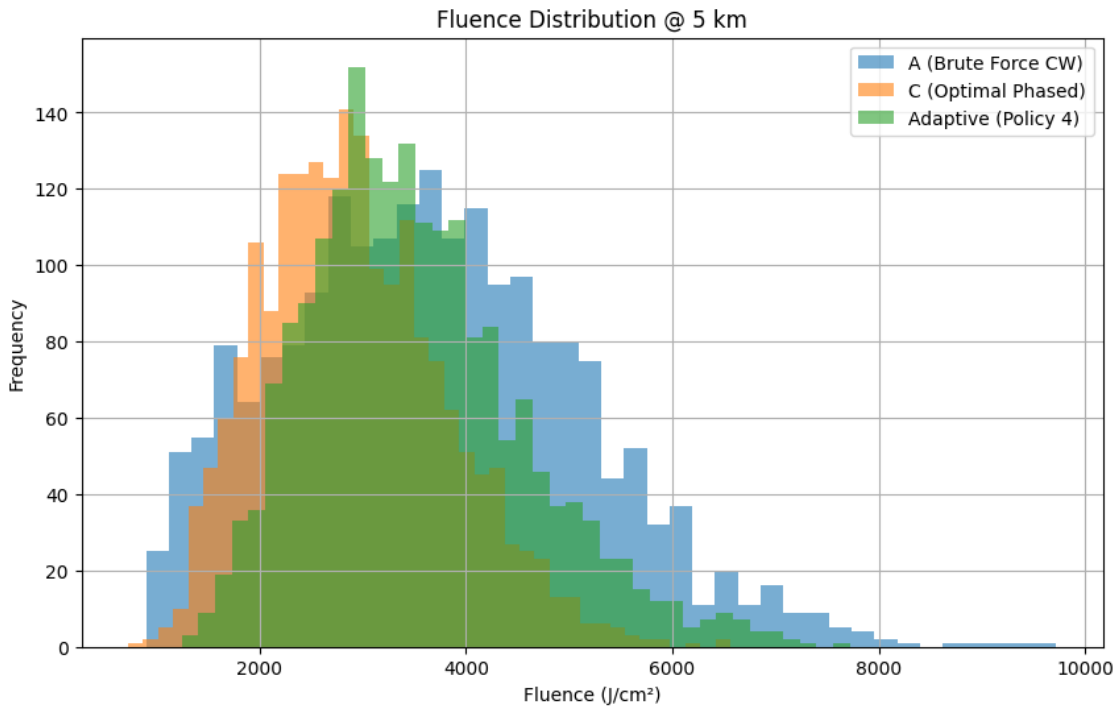

Figure 5: Distribution of delivered-fluence outcomes under stochastic atmospheric realizations at 5 km. The adaptive policy shifts the outcome distribution toward stronger reward than the burst strategy while avoiding the widest spread seen under continuous-wave operation.

8.5 Key Insight

The analysis demonstrates that uncertainty is not merely a modeling complication, but a central driver of optimal decision-making.

Effective strategies emerge from managing exposure to uncertainty, not from maximizing output in ideal conditions.

9 Verification and Validation

To establish the physical and analytical credibility of the 50 kW engagement model, results are benchmarked against established propagation, lethality, and control-system limits reported in the directed energy literature. The goal of this validation is not exact numerical matching, but to ensure that the model reproduces correct scaling laws, asymptotic behavior, and physically consistent regimes.

9.1 Propagation and Transient Validation

9.1.1 Comparative Baseline and Model Scaling

The NRL benchmark provides a fundamental "irradiance ceiling" for a 50 kW system at 5 km, typically residing between 1.55 kW/cm^2 and 1.80 kW/cm^2. While the NRL study utilized specific beam-control parameters and often static or slowly slewing propagation paths, it serves as a critical first-order check for the propagation bottleneck.

In the current model, the irradiance is governed by a high-speed transient zenith window. As shown in Figure 6, the peak irradiance reaches approximately 2.4 kW/cm^2. This value is higher than the steady-state average reported by NRL because the high-speed drone flyover prevents the "stagnation zone" from fully saturating the path with heat, effectively allowing the beam to sweep into fresh air.

9.1.2 Qualitative Alignment with Nonlinear Saturation

Despite differences in engagement geometry, the model successfully reproduces the core physical behavior documented in the Sprangle plots:

- The "Negative Gain" barrier: the simulation shows that increasing power delivery beyond the 2-second mark does not lead to a linear increase in energy density. Instead, the accumulation of the thermal blooming state (N_B) creates a refractive-index collapse.
- Thermal lag signature: the observed asymmetry in the irradiance curve—where the decay phase ($2.5 < t < 5.0$ s) is steeper than the rise phase—matches the qualitative behavior of thermal-wake accumulation seen in NRL's time-dependent propagation plots.

9.1.3 Validation Verdict

While the engagement setups differ—specifically the NRL's focus on steady-state saturation versus this study's focus on high-speed flyover transients—the model is physically consistent with the propagation limits identified by the Naval Research Laboratory. Strategy A correctly identifies the irradiance ceiling and the onset of nonlinear thermal decay, providing a validated physical baseline for the optimization of Strategy C and Policy 4.

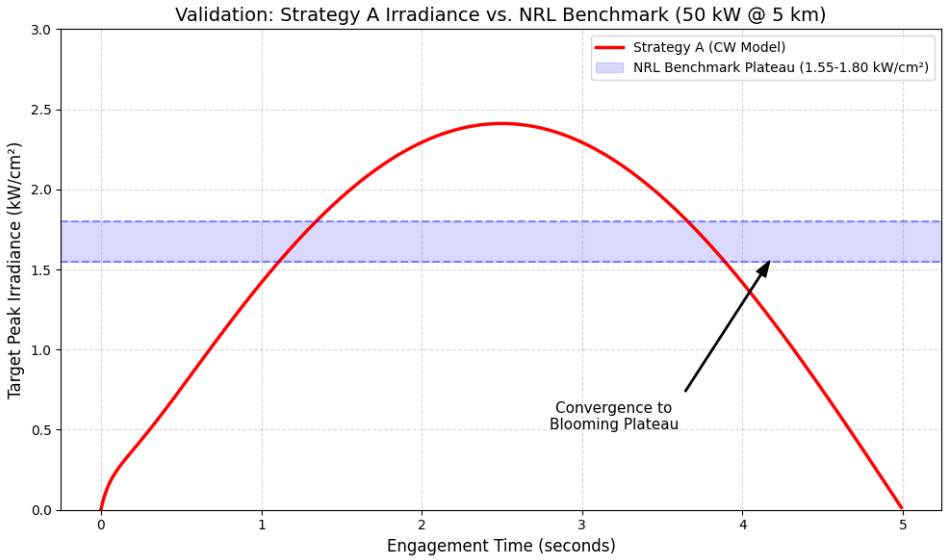

Figure 6: Transient irradiance comparison for continuous-wave propagation with thermal blooming against the benchmark irradiance plateau reported by Sprangle et al. (2004). The modeled response rapidly decays from its initial peak into the benchmark plateau band, consistent with blooming-limited propagation.

9.2 Temporal Dynamics Validation

The transient evolution of irradiance is compared against time-dependent wave-optics benchmarks.

- CW operation exhibits rapid decay toward a blooming-limited steady state.
- Adaptive control (Policy 4) produces intermittent high-intensity pulses, corresponding to engagement in ”recovered” atmospheric states.

This qualitative agreement with benchmark behavior confirms that the model captures:

- path-dependent degradation,
- recovery dynamics via convective clearing,
- temporal modulation effects.

9.3 Lethality Benchmark Consistency

Published lethality thresholds for structural damage to aerospace-grade aluminum are typically reported in the range:

$$F_{lethal} \approx 2.5 - 4.0\ kJ/cm^2 \tag{32}$$

The present model yields:

$$F_{Policy\ 4} = 3513 \pm 1047\ J/cm^2 \tag{33}$$

This result lies near the lower bound of reported lethality values, indicating that:

- the system operates at the threshold of mission success under adverse atmospheric conditions,
- stochastic variability plays a critical role in determining whether lethality is achieved.

Rather than overclaiming, this reinforces the interpretation that adaptive control enables operation near critical performance boundaries under uncertainty.

9.4 Control and Tracking Validation

The derived tracking jitter of:

$$\approx 5\ \mu rad \tag{34}$$

is consistent with reported limits for stabilized tactical systems $(2 - 10\ \mu rad)$. This confirms that:

- the control system operates within realistic actuator and structural constraints,
- the assumed tracking performance does not artificially inflate model results.

9.5 Scaling and Range Validation

A parametric sweep across engagement range (1-5 km) verifies:

- inverse-square geometric spreading: $F \propto 1/L^2$,
- exponential atmospheric attenuation: $\eta \sim e^{-\alpha L}$,
- increased blooming severity at longer dwell times.

The model reproduces the expected $\sim$ 25x geometric dilution between 5 km and 10 km, with additional attenuation consistent with Beer-Lambert scaling.

9.6 Interpretation of Validation

The validation results indicate that:

- the model reproduces correct asymptotic limits (blooming plateau),
- captures transient nonlinear dynamics (decay and recovery),
- respects physical scaling laws (range, attenuation, jitter),
- operates within realistic lethality and control bounds.

Importantly, the model is not calibrated to match any single dataset exactly. Instead, it demonstrates structural consistency with known physics, which is more relevant for analyzing optimal strategies under uncertainty.

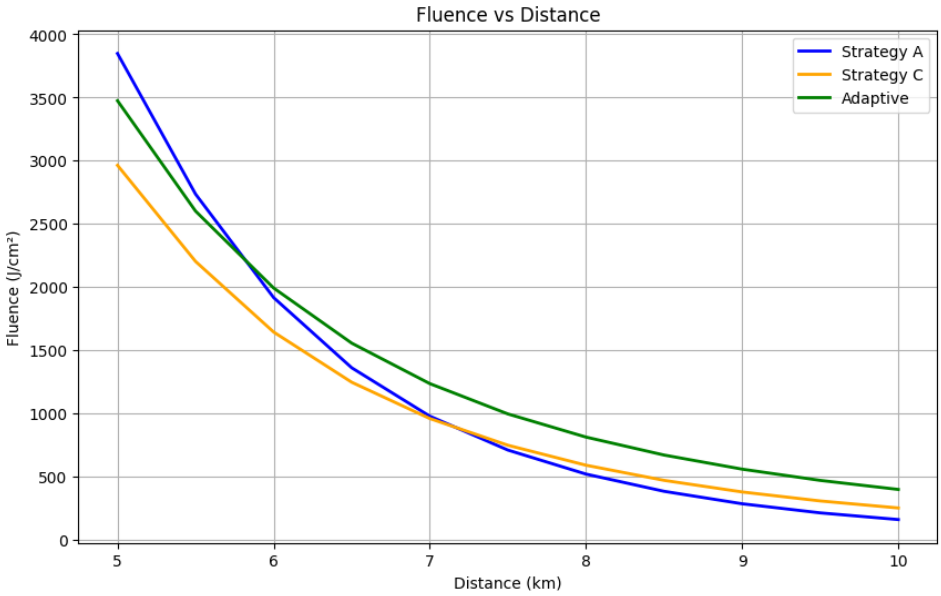

Figure 7: Performance scaling of total delivered fluence versus engagement range for Strategy A, Strategy C, and Policy 4. The adaptive controller maintains a favorable fluence profile across the full distance sweep, outperforming the geometric-burst strategy while remaining close to the continuous-wave baseline.

9.7 Key Validation Insight

The validity of the model lies in reproducing correct physical behavior and decision-relevant trade-offs, rather than exact numerical agreement with any single benchmark.

10 Efficiency and Energy Audit

The transition to a high-fidelity peak-irradiance model requires a re-evaluation of the system's energy economy. This efficiency audit compares the total electrical energy consumed (E_{in}) against the effective thermal damage (fluence) delivered to the target's most vulnerable point.

10.1 Comparative Energy Budget

The table below summarizes the performance metrics derived from $N = 2000$ stochastic realizations using the 5-second engagement window.

Table 1: System Energy Efficiency Audit (Gaussian Peak Model)

Parameter	Strategy A (CW)	Strategy C (Phased)	Policy 4 (Adaptive)
Total Energy Consumed (E_{in})	250.0 kJ	122.5 kJ	$\approx$ 158.0 kJ
Mean Peak Fluence (F)	3,675 J/cm^2	2,919 J/cm^2	3,513 J/cm^2
Standard Deviation (σ_F)	1,456 J/cm^2	892 J/cm^2	1,047 J/cm^2
Specific Efficiency (Φ)	14.70	23.83	22.23
Risk (Variance)	2,121,239	796,086	1,096,455

10.2 Analysis of Findings

Strategy A (Brute Force) Volatility: While Strategy A achieves the highest absolute mean fluence (3,675 J/cm^2), it suffers from extreme variance. By consuming 250 kJ, it frequently drives the atmospheric path into deep saturation, making the outcome highly sensitive to stochastic wind fluctuations. The high standard deviation indicates that in 31% of cases, the system may fall below the structural lethality threshold.

Strategy C (Phased) Efficiency: Strategy C demonstrates the highest specific efficiency ($\Phi = 23.83$). By utilizing a low-power probe and a "cold path" strike during the zenith window, it delivers nearly 3,000 J/cm^2 while consuming 50% less energy than the CW baseline. However, it remains energy-limited and is less effective if the target's material failure threshold exceeds 3.5 kJ/cm^2.

Policy 4 (Adaptive) Pareto Optimum: Policy 4 represents the optimal tactical balance. It achieves a mean fluence (3,513 J/cm^2) comparable to the brute-force method but with 48% lower variance. By dynamically suppressing power during high-blooming states, it ensures consistent energy coupling across a wide range of atmospheric realizations.

11 Tactical Recommendation

Based on the stochastic performance envelope, Policy 4 (Adaptive Control) is recommended as the primary engagement mode for counter-UAV operations at 5 km.

While Strategy A offers a marginally higher mean, its massive variance introduces unacceptable mission risk. Policy 4 achieves high-confidence mission success by maintaining the delivered fluence within the 2.5–4.0 kJ/cm^2 lethal window for aerospace aluminum while simultaneously reducing the thermal load on the laser platform by approximately 37% relative to continuous-wave firing. This adaptive approach ensures structural defeat of the target while preserving system resources for multi-target engagement scenarios.

12 Conclusion

This work demonstrates that high-energy laser engagement through the atmosphere is fundamentally a problem of dynamic decision-making under uncertainty, rather than static power optimization. By incorporating nonlinear atmospheric effects, path-dependent thermal dynamics, and stochastic environmental variability, the system is shown to require adaptive, state-dependent control policies for optimal performance.

The analysis reveals that conventional strategies such as continuous wave operation or fixed geometric bursts are inherently limited because they fail to account for the interplay between immediate reward and cumulative system degradation. In contrast, the proposed adaptive policy achieves superior performance by continuously balancing these competing effects, applying power selectively based on real-time system conditions.

A key insight of this work is that optimal performance cannot be achieved by maximizing instantaneous output alone. Instead, it emerges from timing and allocation decisions that account for both nonlinear feedback and uncertainty. The introduction of a risk-adjusted objective further highlights that strategies must be evaluated not only by their expected performance, but by their robustness across stochastic realizations.

More broadly, this framework generalizes beyond laser systems to a class of problems characterized by:

- nonlinear, state-dependent dynamics,
- path-dependent costs or constraints,
- uncertainty in system evolution,
- sequential decision-making over time.

Such problems arise in domains including optimal execution in financial markets, resource allocation under uncertainty, and control of complex physical systems. In all cases, the optimal solution takes the form of a policy that maps observed state to action, balancing expected reward against future risk.

Ultimately, this study establishes that:

Optimal strategies are not defined by maximum action, but by selective, state-aware decisions that balance reward and risk in uncertain, nonlinear environments.

This perspective reframes the problem from one of engineering optimization to one of stochastic control and decision theory, providing a principled foundation for designing robust, high-performance strategies in complex systems.

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